Chatbot for the Detection of Depression in University Students in Peru Using Sentiment Analysis

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Mental **disorders, such as depression, usually manifest themselves on average by the age of 24, when many young people are in college. During the COVID-19 pandemic, university students experienced a significant impact on their mental health due to confinement, which highlights the need for and importance of early detection of depression so that it can be treated effectively and in a timely manner. Because of this, a chatbot based on sentiment analysis is proposed for the early detection of depression in university students. The algorithm used in the chatbot makes use of responses to the PHQ-9 questionnaire modified to open-ended questions to predict the level of depression that the student may have. This proposal has 4 stages: Technology Analysis, Algorithm Development, Results and Chatbot Construction. The results obtained showed that the Gradient Boosting machine learning technique obtained an accuracy of 0.81. This result suggests that our chatbot can effectively and early detect depression in university students, providing a valuable contribution to the mental health care of students.**

Keywords—chatbot, depression, sentiment analysis, algorithms, NLP, machine learning

I. Introduction

Mental disorders negatively impact the quality of life of people who suffer from them, affecting their personal relationships, their performance at work or in studies. Among the most common disorders is depression which, according to the World Health Organization (WHO), affects approximately 280 million people, including 5% of adults worldwide [1]. Generally, 75% of mental disorders such as depression are usually diagnosed, on average, at 24 years of age, this being an age when young people are studying or are already finishing their university studies [2].

In Peru, studies have been carried out that seek to predict depression through the use of artificial intelligence algorithms, as is the case of Daza et al. [3] which used 3 combined models based on the assembly machine learning algorithm called Stacking to predict depression in university students.

On the other hand, studies have also been conducted focused on the treatment of people diagnosed with depression through the use of chatbots that use virtual reality to combat the shortage of psychologists in universities and help reduce the level of stress in students [4] and chatbots that provide self-help to university students with conversations focused on emotions and empathic responses [5]. However, these chatbots do not focus on detecting disorders early. On the contrary, those chatbots that did detect depression depended on questionnaires with structured Yes and No questions [6] or through Likert scales [3], which limits the patient and does not allow him to express himself openly.

Therefore, our proposal is to implement a chatbot based on artificial intelligence to detect depression early in university students in Peru using natural language processing focused on sentiment analysis, which will have the following phases: (1) Technology Analysis, (2) Algorithm Development, (3) Testing, (4) Chatbot Construction.

The work is organized as follows: In Section 2 the works related to our proposal will be reviewed. Section 3 shows the proposed solution model. Product validation will be done in Section 4. Section 5 will show the results and discussion. The article concludes by showing the conclusions and future work in Section 6.

II. Related Jobs

Chatbots in the field of mental health have shown that they are tools that can streamline clinical processes, automating tasks for the detection and treatment of mental disorders. This section mentions some related jobs that use a chatbot for depression screening.

In the study by Santa-Cruz et al. [6], the Chatbot-together with the abbreviated SRQ questionnaire was used to detect disorders such as depression and anxiety. Users with scores greater than 5 or affirmative answers on questions about suicide or self-harm were contacted by psychologists immediately.

In the study by Kaywan et al. [7], the DEPRA chatbot is proposed that makes use of the SIGH-D and IDS-C structured interview guides for the detection of depression. They chose to turn multiple-choice questions into open-ended questions in order to allow users to share their feelings.

In the study by Abilkaiyrkyzy et al. [8], a dialogue system based on conversational artificial intelligence is proposed to detect mental problems, their severity, and provide personalized feedback. To detect the severity of depression, they used the PHQ-8 questionnaire. In addition, for the development of the chatbot they used the techniques of tokenization, featurizer, classification of intentions and extraction of entities.

III. Chatbot Proposal

In this section, a chatbot focused on the early detection of depression in university students in Peru is proposed. This system will provide an early diagnosis of the level of depression and be able to enable a communication channel with a psychologist to be able to carry out a treatment according to the level of depression diagnosed by the

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chatbot. Therefore, the development of the chatbot will consist of 4 phases (see "Fig. 1").

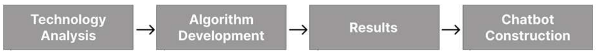

Fig. 1. Phases of chatbot development.

A. Phase 1: Technology Analysis

This phase aims to establish what type of clinical questionnaire focused on detecting depression and what artificial intelligence algorithm will be used to make the diagnosis. Because of this, a comparison of AI algorithms and depression screening questionnaires was developed.

Once the evaluation of clinical questionnaires was carried out, it was determined that the PHQ-9 questionnaire is the most appropriate because it has fewer questions and focuses on measuring both the severity and frequency of symptoms. With regard to AI algorithms, it decided on natural language processing (NLP), due to its great use in chatbot solutions.

B. Phase 2: Algorithm development

This phase aims to develop the algorithm for the detection of depression, so a dataset will be created with which the algorithm will be trained.

The dataset to be created consists of responses to the PHQ-9 questionnaire [9] by 53 university students. The 9 questions have been modified so that they can be answered both openly and through a Likert scale where the values consist of 0: No day, 1 = Several days, 2 = More than half of the days and 3 = Almost every day. The dataset then has the fields: pregunta_abierta and score_likert. Of the 53 students, 47 (87%) demonstrated some level of depression while the other 6 did not have any degree of depression.

```
Start
    Read data from 'data_Training09.csv' into DataFrame df
    Function clean_text(text):
        text = text in lowercase
        text = remove punctuation
        Return text
    Apply clean_text to df['texto']
    Split df into train_data and test_data
    X_train = train_data['texto']
    y_train = train_data['intensidad']
    X_test = test_data['texto']
    y_test = test_data['intensidad']
    Initialize an empty list called results
    For i from 1 to 20 do:
        Create a pipeline with CountVectorizer and GradientBoostingClassifier
        Define param_grid with max_features, ngram_range, n_estimators & learning_rate
        Initialize grid_search with pipeline and param_grid
        grid_search.fit(X_train, y_train)
        best_params = grid_search.best_params_
        y_pred = grid_search.best_estimator_.predict(X_test)
        report = classification_report(y_test, y_pred, output_dict=True)
        Store results in results
    results_df = Convert results to DataFrame
    Save results_df to 'results_classification.xlsx'
End
```

Fig. 2. Phases of chatbot development.

Making use of this dataset, the training of the artificial intelligence algorithm was carried out using natural language processing with sentiment analysis techniques, with the aim of classifying the results in a degree of depression according to the PHQ-9. The "sklearn" library in Python was used where several available machine learning algorithms were identified, such as: Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosting, K-Nearest Neighbors and Naive Bayes.

The code loads a dataset from a CSV file, cleans up the text by removing punctuation marks and transforming it to lowercase, and then splits the data into training (70%) and test (30%) sets. Through a loop of 20 iterations, a pipeline is created that combines a TfidfVectorizer to transform text strings into numeric vectors and a classifier, optimizing hyperparameters using grid search. Subsequently, the performance of the model in the test set is evaluated and a classification report is generated covering four categories of feelings, based on a total of 694 records.

To evaluate the performance of the algorithms, 5 metrics will be used: Accuracy, precision, completeness, F1-Score and area under the ROC curve [3].

C. Phase 3: Results

This phase aims to test and compare the results of the algorithms trained in the previous phase and select the one that has the best performance.

TABLE I. ALGORITHM METRICS COMPARISON

Machine Learning Algorithms	**Algorithm evaluation metrics**				
	Accuracy	***Precision***	***Recall***	***AUC***	***F1-Score***
Logistic Regression	0.70	0.71	0.70	0.80	0.69
Support Vector Machine	0.64	0.66	0.64	0.85	0.64
Random Forest	0.66	0.70	0.66	0.83	0.65
Gradient Boosting	0.70	0.75	0.70	0.83	0.70
K-Nearest Neighbors	0.64	0.63	0.64	0.74	0.62
Naïve Bayes	0.64	0.65	0.64	0.83	0.63

The results obtained are shown in Table I. Regarding Accuracy, the models with the best results were Logistic Regression and Gradient Boosting with 0.70. In accuracy, Gradient Boosting stands out with 0.75. For the completeness or *recall* metric, both Logistic Regression and Gradient Boosting scored 0.70. In the area under the ROC curve (AUC), Support Vector Machine scored the highest with 0.85. Finally, for the F1-Score metric, the model with the highest score was Gradient Boosting with 0.70.

To detect the level of depression of university students, the model that obtained the best scores was Gradient Boosting, especially in the scores obtained in accuracy with 0.75 and F1-Score with 0.70.

D. Phase 4: Chatbot Building

This phase aims to develop the web application where the chatbot will be deployed and validate the correct functioning of the system.

A monolithic project is being developed which integrates, as shown in Fig. 3, the back and the front in a single project, which is consumed through an external API by artificial intelligence. This is because NextJS is used, a full stack framework that allows you to create this type of project, speeding up development, testing and front and back end integration. The Cloud provider chosen to deploy the front is Vercel, which in turn provides us with the PostgreSQL database.

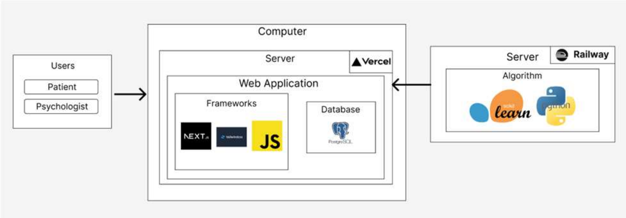

Fig. 3. Logical architecture of the chatbot.

IV. VALIDATION

In this section we will validate the chatbot by performing 3 types of evaluations: validation of the depression detection algorithm, usability of the system and user satisfaction.

The first validation consists of evaluating the accuracy of the Gradient Boosting algorithm. To achieve this, a larger dataset with a total of 156 responses was used. Likewise, the code previously used was modified to obtain better results in relation to accuracy. In turn, the dataset was modified, adding characteristics such as age, career and gender to improve sentiment analysis. Open-ended responses are vectorized using TF-IDF with n-grams of 1 and 2 and the other variables are converted to numbers to balance classes with SMOTE and not cause bias. The dataset is divided into training (70%) and test (30%) and the Gradient Boosting model is trained with optimized parameters.

In the second validation, the usability of the chatbot system will be evaluated through the System Usability Scale (SUS) [8] that analyzes the level of complexity and inconsistencies that may occur during interaction with the chatbot. Use a Likert scale from 1 to 5 where 1 point represents "Strongly disagree" and 5 points "Strongly agree". This questionnaire will be given to a psychologist who specializes in treating university students.

The third validation assesses user satisfaction using the Client Satisfaction Questionnaire (CSQ-8) [5]. This questionnaire will be administered to 15 students of the Information Systems Engineering career of the Peruvian University of Applied Sciences. Prior to taking the questionnaire, the student will have to interact with the system for approximately 5 minutes, browsing all the screens and finally solving the PHQ-9 questionnaire with the help of the chatbot. Once the interaction is finished, they will answer the CSQ-8 questionnaire.

V. RESULTS AND DISCUSSION

Table II shows the results obtained based on the new training of the Gradient Boosting model:

TABLE II. SELECTED ALGORITHM METRICS

Selected Algorithm	Algorithm evaluation metrics	
	Precision in training	*Validation accuracy*
Gradient Boosting	1.0	0.81

Compared to the results shown in Table I, the accuracy obtained for Gradient Bossing in this new workout increased significantly from 0.75 to 1.0. This increase in accuracy is attributed to the addition of characteristics such as gender, age and university degree and also to the large increase in records in response that was obtained, thus increasing the information included in the dataset used. The balancing of classes and the use of n_gramas 1 and 2 were also included.

Comparing the results obtained with other studies, the study carried out by Daza et al. [3] where, with a dataset of 284 students, Gradient Boosting was used to classify depression and an accuracy of 79.48% was obtained. Our model scored slightly higher in terms of accuracy even though this study has a slightly larger dataset and also employed class balancing. It should be noted that the authors only used the PHQ-9 Likert scale, so an important factor in the result of our metrics may be due to the inclusion of open-ended responses.

On the other hand, Qu et al. [10] used eXtremeGradientBoosting (XGBoost) to predict depression in 2546 U.S. veterans and obtained an accuracy of 0.942. This high accuracy score can also be attributed to the large number of samples for the dataset as they used the National Health and Nutrition Examination Survey (NHANES) database.

The usability of the system was evaluated through a form sent to an expert in clinical psychology after visualizing the flow of patient and psychologist users in the system. Table III shows the score given by the psychologist to each of the items of the SUS questionnaire. The final score of the questionnaire was 80, which is above the average score for this questionnaire which is 68 [8]. The items with the highest scores were items 5 and 7, which talk about the integration of components and the ease of learning the use of the system, respectively.

According to the results of the user satisfaction validation, taking into account that the minimum score that can be obtained is 8 and the highest 32, the average score obtained is 26.13. Of the 15 students, 7 scored below average, however, 3 students scored the chatbot with the maximum score. These results show that there is an interest in using the chatbot for the early detection of depression, however there are still areas for improvement to obtain higher satisfaction scores from our patient users. In Table IV, it can be seen that question 5 is the one with the lowest score on average of the entire questionnaire.

TABLE III. RESULTS OF THE SUS QUESTIONNAIRE

Items	Pts
I think I'd like to use this system often	4
I found the system unnecessarily complex	2
I thought the system was easy to use	4
I think I would need the support of a technician to be able to use this system	2
I found that the various functions of this system were well integrated	5
I thought there was too much inconsistency in this system	2

Items	Pts
I imagine that most people would learn how to use this system very quickly	5
I found the system very complicated to use	2
I felt very confident using the system	4
I needed to learn a lot of things before I started with this system	2

TABLE IV. CSQ-8 QUESTIONNAIRE RESULTS

Items	Prom.
How would you rate the quality of the service you received?	3.20
Did you get the kind of service you wanted?	3.33
To what extent has our program met your needs?	3.27
If a friend needed similar help, would you recommend our program?	3.53
How satisfied are you with the amount of help you received?	3.00
Did the services you received help you deal more effectively with your problems?	3.27
Overall, how satisfied are you with the service you received?	3.2
If you needed to seek help again, would you return to our program?	3.33

VI. CONCLUSIONS AND FUTURE WORK

In conclusion, the chatbot allowed us to predict the level of depression in university students using artificial intelligence algorithms. The chatbot was evaluated against predetermined metrics such as accuracy, precision, completeness, F1-Score, and area under the curve (AUC). Likewise, the usability of the system and user satisfaction in general were evaluated.

The algorithms proposed to detect depression and be integrated into the chatbot were evaluated based on metrics such as accuracy, precision, completeness, F1-Score and area under the curve (AUC). It was shown that the Gradient Boosting algorithm is the one that demonstrated the best performance in the detection of depression in university students, obtaining 81% in the test phase.

The expert's validation shows us that the chatbot based on artificial intelligence is a good tool for detecting depression and that it is easy to use due to the good integration of its components. Regarding user satisfaction evaluated in 15 university students, it was found that 81.6% have an interest in using the system as a tool to help early detection of depression.

As a future work, it should be applied in state universities and in this way expand the sample of the dataset to obtain better results as well as obtain longer and more complete answers to each of the questions. On the other hand, it would also be interesting to compare the results obtained from different university degrees.

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